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SUPERSEDING
NAVORD OS 5950B
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MILITARY SPECIFICATION

MICROPHONE MARK 10 MOD 0

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of Microphone Mk 10 Mod 0, and the methods of inspection and tests upon which acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621	-- Boxes; Wood, Nailed and Lock-Corner.
LLL-B-631	-- Boxes; Fiber, Corrugated (For Domestic Shipment).
PPP-B-601	-- Boxes; Wood, Cleated-Plywood.

MILITARY

JAN-P-106	-- Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
MIL-P-116	-- Preservation, Methods of.
MIL-B-131	-- Barrier Material; Water-Vaporproof, Flexible.
MIL-C-17217	-- Crystal Plate, ADP (For High-Power Sonar Transducer).

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STANDARDS

MILITARY

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 275882 -- Microphone Mk 10 Mod 0,
LD 276066 -- Test Set Mk 207 Mod 0,
491127 -- Hydrophone Mk 5 Mod 0 Assembly,
491143 -- Hydrophone Housing,
and all documents listed on the above LDs and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the Procuring Activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 Microphone Mk 10 Mod 0 covered by this specification shall be manufactured in accordance with Bureau of Ordnance List of Drawings LD No. 275882 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Crystal plates. -- The crystal plates shall be homogeneous and shall be free from flaws, cracks, and chips as defined in Military Specification MIL-C-17217.

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3.3 Pressure tightness. -- Each microphone assembly shall be pressure tight so as not to leak either water or oil when subjected to a water pressure of 200 pounds per square inch on the diaphragm side. The Procuring Agency will conduct tests to insure that the quality of materials and assembly is maintained to guarantee conformance with this requirement.

3.4 Sensitivity. -- The sensitivity of the microphone shall be judged on the basis of a prototype Microphone Mk 10 Mod 0 to be supplied by the Procuring Agency which microphone shall serve as the primary standard for all tests. The sensitivity of each microphone when tested as specified in 4.4.1 shall not differ from the sensitivity of the primary standard by more than +30 percent, -20 percent for frequencies up to 700 cycles per second, and shall also have a minimum output of 10 db below that of the primary standard at the frequencies of 5000 and 10,000 cycles per second.

3.5 Frequency response. -- The frequency response requirement, tested as specified in 4.4.2, is a design requirement and shall not be used as a basis of acceptance or rejection of the microphone. However, this requirement is an important index of quality and the test shall be performed on the preproduction sample and on any other samples whenever considered necessary by direction of the Procuring Agency.

3.6 Freezing. -- After being exposed to the freezing test specified in 4.4.3 the sensitivity of the microphone shall not differ from the sensitivity of the primary standard by more than ±25 percent when tested as specified in 4.4.1.1.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Standard MIL-STD-105.

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection

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Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of twelve (12) Microphones Mk 10 Mod 0 shall be delivered by the Contractor for test to an agency designated by the Procuring Agency. These microphones shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provisions of General Procedures for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.4 Test Procedure.

4.4.1 Sensitivity tests. -- The procedure for the sensitivity tests shall be governed by the method of manufacturing employed for Microphone Mk 10 Mod 0. If a single Contractor undertakes to build the microphone as a complete unit at one location, the procedure shall be as specified in 4.4.1.1 and 4.4.1.2. If the crystal assembly

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and preamplifier are manufactured separately, the procedure shall be as specified in 4.4.1.3.

4.4.1.1 Test procedure when a complete microphone is manufactured. Each Microphone Mk 10 Mod 0 shall be tested by the substitution method specified in 4.4.1.1.1 through 4.4.1.1.6.

4.4.1.1.1 Tests shall be conducted by placing the microphone concentrically upon the face of the transducer assembly from a Test Set Mk 207 Mod 0, Bureau of Ordnance List of Drawings LD No. 276066. This test set and transducer assembly shall be supplied by the Procuring Agency. Precautions shall be taken to insure vibration isolation between the transducer and the supporting medium.

4.4.1.1.2 The microphone output shall be measured by means of the output meter of Test Set Mk 207 Mod 0.

4.4.1.1.3 The test frequencies shall be those corresponding to positions 1 through 4 of the oscillator selection switch of Test Set Mk 207 Mod 0. These frequencies are approximately 180, 300, 450, and 700 cycles per second.

4.4.1.1.4 All microphones shall be stored for at least one week before testing.

4.4.1.1.5 During a period of four hours prior to testing, each microphone shall be stored in an atmosphere which shall not differ by more than 3 degrees C from the temperature at which calibration is made and which shall be maintained in the range of 20 degrees to 30 degrees C.

4.4.1.1.6 Any microphone which indicates a sensitivity which differs from that of the primary standard by more than +30 percent -20 percent at any of the test frequencies shall be rejected.

4.4.1.2 In addition to testing the microphones as specified in 4.4.1.1, the microphone shall also be tested as specified in 4.4.1.2.1 through 4.4.1.2.5.

4.4.1.2.1 Each microphone selected for this test shall first have its preamplifier separately tested and accepted as specified in 4.4.1.3.2.

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4.4.1.2.2 The microphone shall be tested in water by a substitutuion method. In the case where a water tank may be used, precautions shall be taken to prevent reflections from its sides. The distance between the microphone and the transducer shall be large enough to prevent formation of standing waves at all test frequencies.

4.4.1.2.3 The microphone shall be mounted in a hydrophone housing, Bureau of Ordnance Drawing No. 491143, which will be provided by the Procuring Agency for these tests. Proper assembly is shown on Bureau of Ordnance Drawing No. 491127.

4.4.1.2.4 A Test Set Mk 207 Mod 0 without the transducer assembly shall be used in this test to supply power to the microphone and monitor the output. Means shall be provided to drive acoustically the microphone at frequencies of 5000 and 10,000 cycles per second. The magnitude of these signals shall be such as to produce 0.75 volt output of the standard microphone at the distance chosen for the test, and for each test frequency.

4.4.1.2.5 Any microphone whose output at these frequencies is more than 10 db below that of the primary standard shall be rejected.

4.4.1.3 Test procedure when crystal assembly and preamplifier are manufactured separately. -- The crystal assembly and preamplifier shall be tested independently as specified in 4.4.1.3.1 and 4.4.1.3.2.

4.4.1.3.1 Substitute test for each microphone crystal assembly.

4.4.1.3.1.1 A standard preamplifier which will be furnished by the Procuring Agency shall be connected to the crystal assembly.

4.4.1.3.1.2 The entire test procedure for the complete microphone as specified in 4.4.1.1 shall apply.

4.4.1.3.2 Test for each preamplifier.

4.4.1.3.2.1 Test Set Mk 207 Mod 0 without the transducer assembly shall be used in making these tests.

4.4.1.3.2.2 The a-c input and output voltages of the preamplifier shall be measured by means of Test Set Mk 207 Mod 0.

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4.4.1.3.2.3 The gain at each test frequency shall be measured at a signal level of a-c output of the preamplifier of 1.0 volt rms.

4.4.1.3.2.4 The test frequencies to be used shall be all eleven frequencies of the test set.

4.4.1.3.2.5 Any preamplifier whose sensitivity differs from the values given in Table I by more than the stated tolerances at any of the test frequencies shall be rejected.

Table I.

Frequency (Cycles per Second)	Gain (Percent)*
180	160 ± 15
300	160 ± 15
450	155 ± 15
700	150 ± 15
1600	155 ± 15
3000	192 ± 15
4000	230 ± 15
6000	290 ± 20
9400	375 ± 20
11,500	400 ± 20
13,500	410 ± 20

* The gain of the preamplifier is defined as the ratio of the a-c output voltage to the a-c input voltage.

4.4.2 Frequency response. -- The following test shall be performed on the preproduction sample and on any other samples when directed by the Procuring Agency.

4.4.2.1 Each microphone selected for this test shall first have its preamplifier separately tested and accepted as specified in 4.4.1.3.2. In addition, the frequency response curve for this preamplifier shall be provided to the Procuring Agency.

4.4.2.2 The microphone shall be tested in water by measuring its output voltage at each test frequency for constant sound pressure at the microphone. In the case

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where a water tank may be used, precautions shall be taken to prevent reflections from its sides. The distance between the microphone and the transducer shall be large enough to prevent formation of standing waves at all test frequencies.

4.4.2.3 The microphone shall be mounted in a hydrophone housing, Bureau of Ordnance Drawing No. 491143, which will be provided by the Procuring Agency for these tests. Proper assembly is shown on Bureau of Ordnance Drawing No. 491127.

4.4.2.4 A Test Set Mk 207 Mod 0 without the transducer assembly shall be used in this test to supply power to the microphone and monitor the output. Means shall be provided to drive, acoustically, the microphone at frequencies of 180, 300, 450, 700, 1600, 3000, 4000, 6000, 9400, 11,500, and 13,500 cycles per second. The magnitude of the sound pressure for each test frequency shall be constant.

4.4.2.5 A frequency response curve for the complete microphone when tested as specified in 4.4.1 shall be provided the Procuring Agency. The sound pressure standard and the value of sound pressure at the microphone shall be included with the curve.

4.4.3 Freezing test. -- Microphone Mk 10 Mod 0 assembly shall be subjected to the following freezing test. Place the microphone in a refrigerated atmosphere, the temperature being held at 0 degrees F or lower, and allow it to remain there for at least 12 hours. Next, place it diaphragm downward in about three inches of water which shall be maintained at a temperature in the range of 45 degrees F to 90 degrees F. Allow the microphone to remain in the water for two hours or until the temperature of the microphone is above the dew point of the surrounding atmosphere. Care shall be taken to keep all water off all parts of the preamplifier on the back of the microphone assembly.

5. PREPARATION FOR DELIVERY

5.1 Packaging (domestic shipment). -- Each microphone shall be indiyidually packaged in a corrugated fiberboard container in accordance with the requirements of Specification LLL-B-631. The microphone shall be supported in all positions by the microphone housing flange and guard-ring, with

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sufficient fillers of the same material. A filler pad shall be positioned, in a compact manner, between the mounting plates of the preamplifier to reduce vibration during shipment.

5.2 Packaging (overseas shipment). -- Each microphone, packaged as specified in 5.1, shall be packaged Method IIe (less desiccant) in accordance with Specification MIL-P-116. Barrier material shall conform to the requirements of Type 1, Class A, Specification MIL-B-131.

5.3 Packing (domestic shipment). -- Microphones, packaged as specified in 5.1 shall be packed in groups of four in a wood-cleated plywood or nailed wood box conforming to the requirements of Specifications PPP-B-601 or NN-B-621, respectively.

5.4 Packing (overseas shipment). -- Microphones, packaged as specified in 5.2, shall be packed in groups of four in a wood-cleated plywood box or a Style 2, 2 1/2, 3, or 4 nailed wood box conforming to the requirements of Specifications PPP-B-601 or JAN-P-106, respectively.

5.5 Marking (domestic and overseas shipment). -- In addition to any special marking required by the contract or order, all containers shall be marked in compliance with Standard MIL-STD-129. The notice, "FRAGILE INSTRUMENTS - HANDLE WITH CARE" shall be prominently stencilled or stamped on the shipping container.

6. NOTES -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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